InterDev

Evaluation Protocol

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Purpose

To evaluate how three different interface designs (Baseline Search, LLM Chat, LLM Chat + Knowledge Graph Citations) support users in retrieving, interpreting, and synthesizing international development evidence within curated collections. We measure task performance, verification behaviour, reliance, workload, subjective impressions, trust, and perceived accuracy across interfaces.

Setting

The study will be conducted online inside the Qualtrics survey platform, with external task execution running in the InterDev system. Participants complete the study remotely on their own devices. The study session is expected to take approximately 40 minutes, though timing may vary depending on the user's pace. All instructions, task transitions, and subjective assessments are delivered via Qualtrics, while retrieval/synthesis/validation tasks occur within embedded links to InterDev.

Central Research Question

Below is the central research question for this thesis. Please keep this in mind throughout the study.

> *"How does adding explicit source/citation surfacing (via a knowledge graph) to an LLM chat interface affect users' ability to retrieve, synthesize, and verify development evidence, and how does it change trust, perceived accuracy, workload, and reliance compared to chat without sources?*

Flow Overview

Each participant will use all three interface variants, but the order in which they encounter them will be counterbalanced using a Latin Square design. This ensures that each condition appears equally often in the first, second, and third block positions across participants, mitigating order effects. In addition, task order is balanced independently, such that each of the three task-pairs appears in each block position across the study sample.

- **Sequence A: C0 → C1 → C2**
- **Sequence B: C1 → C2 → C0**
- **Sequence C: C2 → C0 → C1**

The balancing and distributing of conditions across participants is entirely handled by Qualtrics, you don't need to think about it.

Introduction & Demographics (6 questions)

Participants begin with a short introduction outlining the purpose and structure of the study. This is followed by demographic and background questions capturing domain exposure, familiarity with large language models, and related attributes.

1. **Interface Sequence**

Participants then complete three evaluation blocks. Each block corresponds to one of the following interface conditions:

Each interface sequence consists of:

- **Tutorial Task (1 example task preceded by a short demo by the facilitator):** A low-stakes practice task to allow participants to become familiar with the interface before the main evaluation.
- **Task Notification:** A reminder that the upcoming tasks are part of the formal evaluation and should be attempted carefully.
- **2 Experimental Tasks:** Participants complete two tasks within the same policy domain theme (fixed ordering inside the pair). Each pair is unique per block, ensuring all participants complete all six tasks across the study, but never repeat a task.
- **NASA-TLX Workload Assessment (6 sliders):** After each block, participants complete a short assessment using NASA-TLX and brief trust / verification questions.
- **Trust Scale:** After each block, participants complete a short assessment using NASA-TLX and brief trust / verification questions.

2. **Post Evaluation Discussion and PSSUQ**

Following the completion of all conditions, the participant completes a **Post Study System Usability Questionnaire (PSSUQ).** This is a 19 question Likert scale measure. It is built into Qualtrics, the participant can complete it themselves.

Following this participants participate in a short interview about their experiences with the interfaces. This includes preference, perceived trustworthiness, task-interface fit, perceived accuracy, and how the presence or absence of surfaced citations influenced verification behaviour. The questions will appear on Qualtrics, you will be the one to ask the questions and guide the conversation.

Facilitator Instructions – Study Introduction

This study is run synchronously over Zoom with live facilitation. Before beginning, greet the participant, confirm their audio and video connection are working, and explain that you will guide them through each stage of the study. Participants will complete all tasks on their own machine, in their own browser environment.

Provide the participant with the Qualtrics survey link:

Ask them to open the link in a modern desktop browser (Chrome recommended) and maximise the browser window.

Participants will be asked to complete the consent form before starting the survey. They will be asked to upload their completed consent form here before continuing. Participants should complete the Qualtrics sequence independently; the facilitator is present to clarify instructions and procedural questions, but not to intervene in the task execution.

Participants will be required to navigate to InterDev themselves during the study (when instructed by the survey). They should open a second tab for:

The participant will switch between Qualtrics and InterDev as needed. The facilitator should confirm that the participant is able to access InterDev before beginning the first tutorial. Participants should share their screen during the session. This enables the facilitator to observe how they are interacting with the system for research validity purposes (e.g., where they navigate, what verification behaviour they exhibit, if clarification is needed).
If the user accidently refreshes or closes the Qualtrics tab, it will automatically reopen where you left off if you return to the survey via the link.

Example Introduction

"Hello, and thank you for joining. Before we begin, I'd like to briefly explain how this session will run. You will be completing this study on your own computer, and I will be here the entire time to guide the process, answer questions, and provide clarification when needed."
"You'll have two browser tabs open during the session: the Qualtrics survey, which will guide the experimental flow, and the InterDev system, which you will use to carry out the tasks. The survey will instruct you when to switch between the two."
"Please open the Qualtrics survey now. You can upload your completed consent form onto this page. If you haven't yet filled it out please download and do so now. Participation is voluntary and you can stop at any time. Your data belongs to you: if you want it to be deleted at any time you simply need to let us know"
"In this study, you'll be interacting with three different versions of an evidence synthesis interface for international development and policy research. Each interface offers a different type of assistance. For each one, you'll complete a short tutorial, then two evaluation tasks, and then a short questionnaire about your experience using that interface. The session should take approximately 40 minutes in total."
"If anything is unclear at any point, just let me know. You can ask clarifying questions at any time."
"Before we start, can you please confirm that you can see the Qualtrics survey clearly on your screen, and that you are able to also open a second tab for interdev.adaptcentre.ie when prompted?"

Pre-Study Questionnaire

At the beginning of the Qualtrics survey, participants will be presented with a short questionnaire that collects background information. They will be asked to provide a pseudonym to identify them for the purposes of the study. This includes questions about their familiarity with international development evidence (e.g. reading reports, working with evaluations), their familiarity with large language models, and their awareness of online repositories or platforms that provide development research or evidence summaries. This is followed by basic demographic information such as age, gender, and first language. Introduce this section by saying:

"Before we begin the main part of the study, the survey will ask you a few short questions about your background, including your familiarity with development research and use of AI tools, as well as some demographic information. These questions help us contextualise the results and understand differences between people's prior knowledge and experience. These are standard research questions and will only take a few minutes. Please answer them honestly. You may select 'Prefer not to say' for any question where you would rather not provide a response. Once you complete this section, the survey will automatically guide you

to the first interface tutorial. You will also be asked to choose a pseudonym to identify you for the purposes of the study, which we ask you to use again when logging into the system later"

Ensure that the participant understands that these questions are for research context only, that their responses are confidential, and that they may skip any question they do not wish to answer (e.g., "Prefer not to say" options).
NB Once the Pre-Study Questionnaire is complete please advise the participant that you will be now turning on the recording on Zoom and ensure that they are aware and comfortable with this. Then don't forget to turn it on!

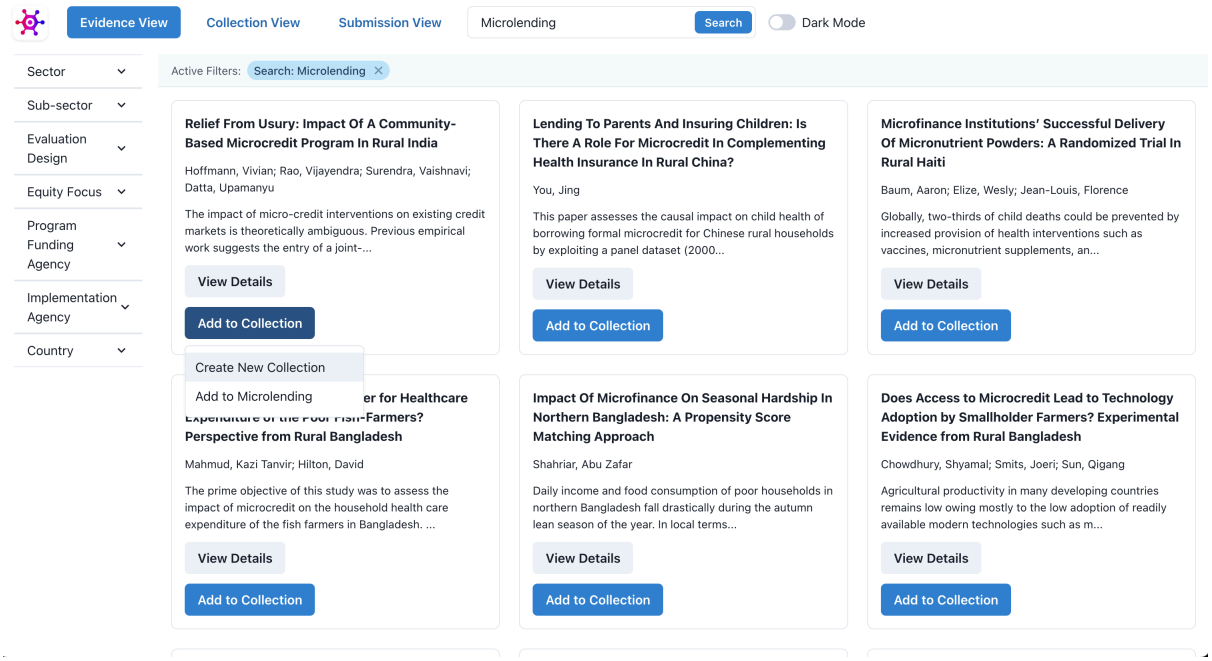
Facilitator Pre-Session Checklist

1. **Zoom setup**
    - Confirm audio and video are working.
    - Ask the participant to share their screen – the web browser is sufficient.
2. **Consent**
    - Ensure the consent form is uploaded before proceeding.
3. **InterDev**
    - Go to the InterDev website www.interdev.adaptcentre.ie
    - Ensure the interface is loading correctly. Create a quick collection, go to the collection view and make sure the chat function is working.
    - Verify it responds correctly and is ready to switch to when needed.
    - Log out when you are ready (cmd-k/ctrl-k and then the button in the top right)
    - Check each interface is working correctly (cmd-k/ctrl-k to switch)
    - When the user first opens Interdev, the will be asked to "log in". **Make sure they use the same pseudonym for this as they did in the demographic survey.**
4. **Final confirmation**
    - Check that both Qualtrics InterDev are working.
    - Be ready to record on Zoom once the tasks begin
    - Be ready to answer participant questions before tasks begin.

Facilitator Instructions – General Introduction

Purpose of this section
Provide the participant with a general introduction of the InterDev system. This is relevant for all conditions of the interface.

Evidence View

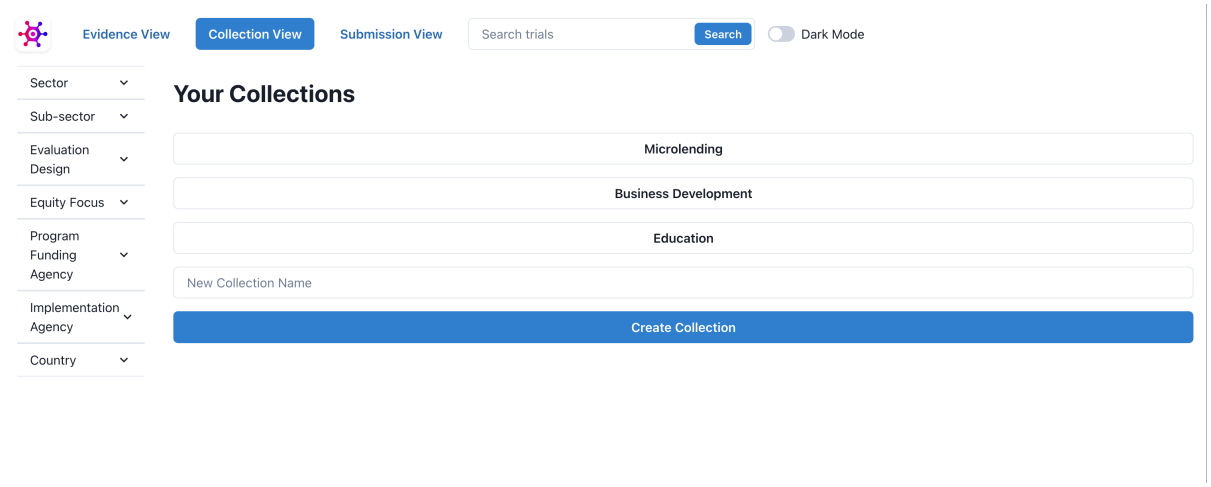


The evidence view is where most individuals will begin their interaction with the system. All trials in the knowledge graph are represented here. Users can search the trials using the searchbox on the top navbar, or do a faceted filter using the navbar on the left side of the screen.

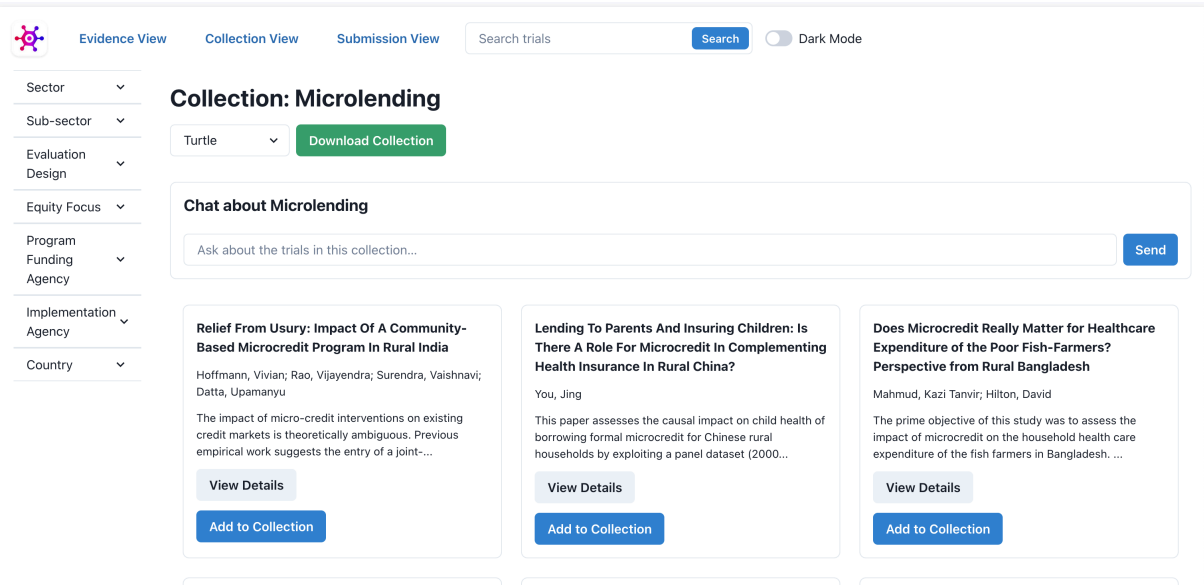
On this screen, users can add trials to a given collection using the "Add to Collection" button. This will also allow them to create a new collection. They can click on the View Details button to learn more about the trial (abstract, authors, country, sector etc).

Using the Collection View and Submission View buttons at the top of the screen, users can navigate to different views for the interface. For this evaluation, we are just using the **Evidence View** and **Collection View**. We don't need to worry about the submission view this time.

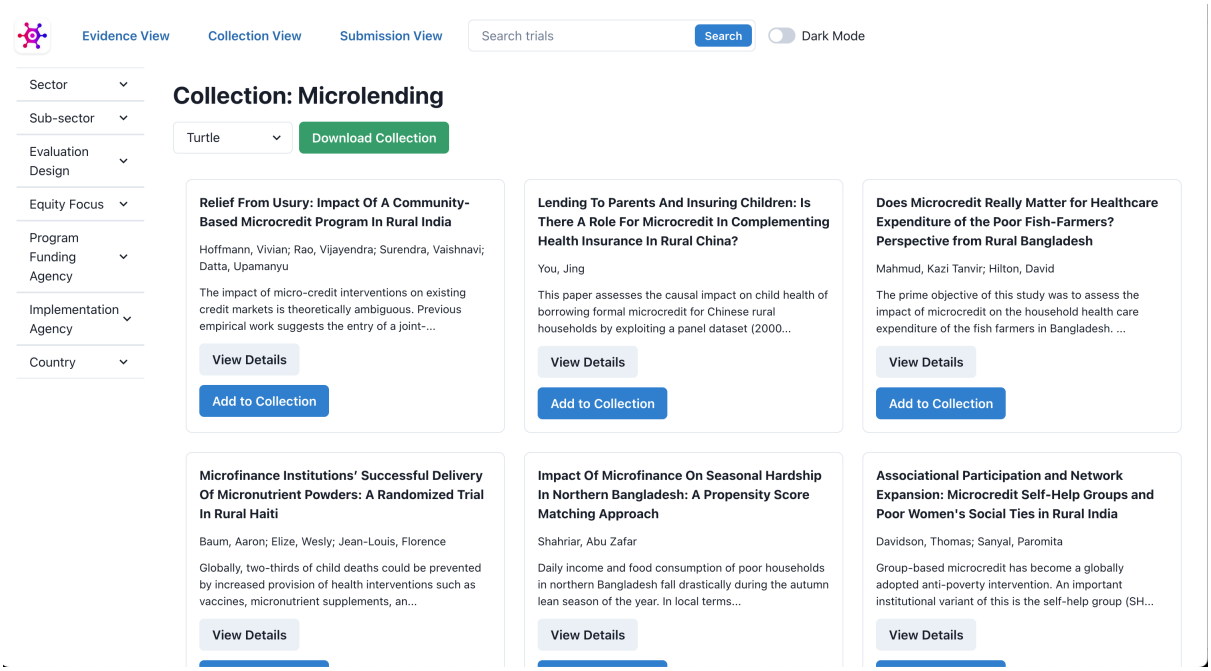
Collection View



The collection view is where all of the collections of trials that the user has created live. These are populated from the evidence view. You can make a new collection here and click to view existing collections (eg Microlending, Business Development, Education above).



Clicking on a collection will give the view above. This is where most of the evaluation will be focused on, as the three conditions vary this view. The baseline condition (below) will not have the chat bar. The chat and chat + sources condition (above) will.
On this screen, users can explore the trials in the collection and use the same filtering tools as the evidence view. If the chat bar exists, they can use this to learn more and synthesise the data within the collection.



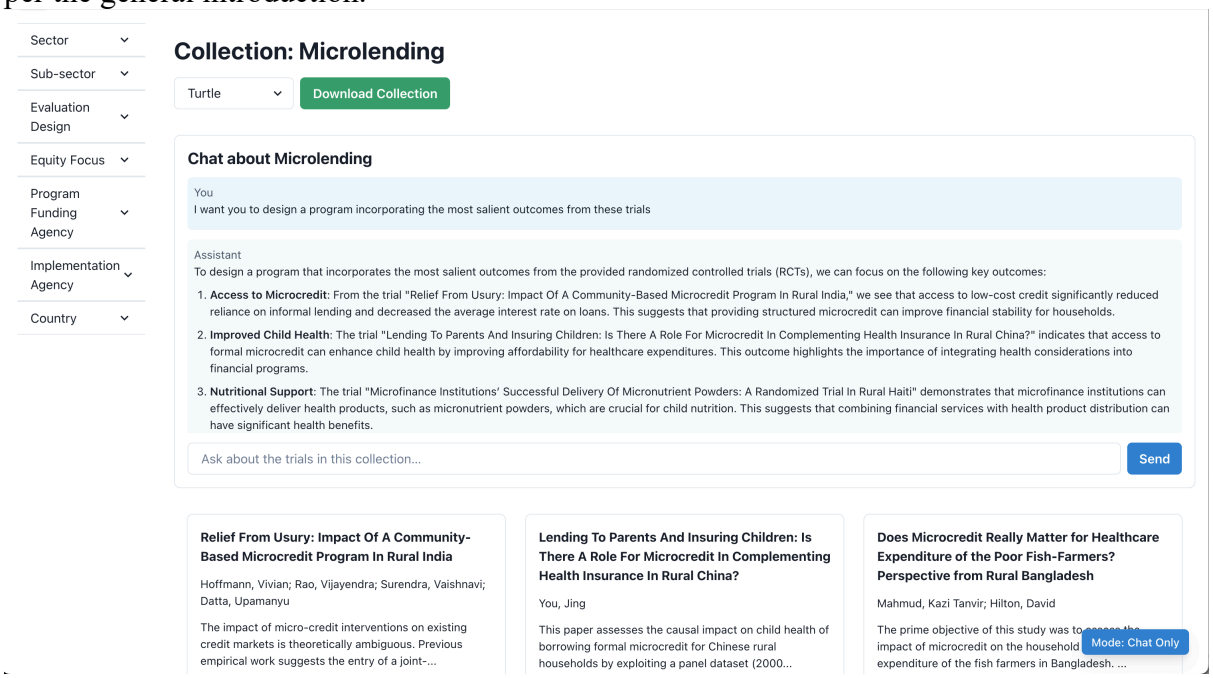
Outside of the tutorial, DO NOT INTERFERE OR HELP the participant if they are stuck. Encourage them to finish the task themselves and if they absolutely cannot figure it out to move on to the next one.

Facilitator Instructions – C0 Condition (Baseline)

Purpose of this section
Introduce participants to the baseline version of the interface. This is as exactly described in the general introduction above – no further introductions necessary.

Facilitator Instructions – C1 Condition (Chat)

Purpose of this section
Introduce participants to the chat interface and ensure they understand how to query and interpret results before attempting the formal tasks.
To introduce the participant to this version of the interface, simply give them an example of using the chat in the collection view, such as that in the image below. Everything else is as per the general introduction.



Facilitator Instructions – C2 Condition (Chat + KG References)

Purpose of this section
Introduce participants to the chat interface and ensure they understand how to interact with it effectively before attempting the formal evaluation tasks.
There is basically no functional difference in this condition versus the condition with chat only, only that the chat will reference the knowledge graph in its response. So, to introduce the participant to this version of the interface, simply give them an example of using the chat in the collection view again. Everything else is as per the general introduction and chat introduction.

Post-Study Interview and PSSUQ

Directly after completing all of the conditions, the participant will complete a Post Study System Usability Questionnaire (PSSUQ). They will complete this themselves in Qualtrics.

Following this, participants will complete a short post-study interview. This section asks them to reflect on their experience with the **three interface variants**. The questions focus on their preferences, how well each interface supported reasoning, synthesis, and comparison,

whether they felt the need to verify claims, and how trustworthy they judged the information provided across interfaces. The questions are below. They will also appear on the Qualtrics survey.

Question Prompt	ID
Of the three interfaces you used (Baseline, Chat, Chat + KG Citations), which did you prefer overall, and why?	Q1
Which interface felt most supportive when you needed to compare or combine findings across multiple evaluations or sources?	Q2
During the tasks, were there particular moments where you felt especially confident in the answer you were producing? Please describe.	Q3
How did the presence or absence of AI support (and citation visibility) influence how much you trusted the information presented?	Q4
Thinking about cognitive effort, which interface felt easiest to work with, and why?	Q5
Is there anything you wanted the system to do that none of the interfaces currently supported?	Q6

Introduce this section by saying:

"You've now completed all of the tasks. To finish, we would like you to answer a short set of reflection questions about your experience with the three interfaces. These questions ask about your preferences, how well you felt each interface supported the work, and your impressions of trustworthiness and verification. There are no right or wrong answers — we are interested in your personal perspective. Please take your time and answer as carefully and thoughtfully as you can."

This section may also be treated as a short semi-structured conversational interview, if appropriate. The facilitator can prompt or follow up on specific parts of the participant's comments where relevant to trust, verification behaviour, or interface comparison, but should not reinterpret or guide their answers.
"That brings us to the end of the session today. Thank you very much for taking part. Your contributions are extremely valuable to this research. We appreciate the time and effort you've put into working through the different systems and tasks. Your responses will help us better understand how people retrieve, evaluate, and synthesise development evidence."
"Just a reminder that your participation is completely confidential, and if you would like to withdraw your data, you can contact us within and your responses will be removed. If you have any questions about the study, please feel free to reach out to us."

Compensation Form Email

After completing the evaluation, please send an email with the following form so that we can compensate participants (usually within a month).

https://scsstcd.qualtrics.com/jfe/form/SV_7V5CRiPaaD4yrXg
Subject: Compensation Form for Study Participation
Dear [Participant's Name],
Thank you again for taking part in our study. To process your compensation, please complete the short form at the link below:
Compensation Form
Once submitted, the gift voucher is usually processed within a month.

If you have any questions or run into issues with the form, please don't hesitate to let me know.

Best regards,
[Your Name]

No Show Email

Hi [Participant Name],

Sorry to have missed you today. If you'd still like to take part in our study, you can reschedule at a time that works for you using this link: [Insert Scheduling Link].

We'd be glad to have you participate.

Best regards,
[Your Name]

When the study is completed and you have received the Zoom recording link, please update the following sheet: